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**Anaphora resolution and reanalysis during L2 processing:
A replication study using the visual world paradigm.**

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Abstract

This bachelor's thesis project focuses on second language acquisition and anaphora resolution. Anaphora resolution is the phenomenon where a piece of language makes a connection with a previously mentioned item in a discourse, for example: "Joseph spoke to John, he was very happy", here the pronoun "he" is acting as an anaphora because it refers to one of the firstly mentioned people in the sentence. Anaphora resolution is a deeply important research field since it reveals how humans process pragmatic connections in between utterances. A psycholinguistic experiment to investigate how advanced L1 Spanish – L2 English bilinguals process anaphora resolution is proposed. It has been seen that different languages have different strategies to process this phenomenon. Our study centres itself on how the two languages interact with each other and whether they may affect each other on this matter. Interestingly, it has been observed that Greek people learning English are able to acquire and properly apply the English anaphora resolution strategy when speaking English (Cunnings et al. 2017). Therefore, our question is then: Do Spanish people learning English acquire the English anaphora resolution strategy or do they use the Spanish strategy to process anaphora in English? Additionally, we asked ourselves: If a noun, that contradicts an initial assessment, is presented will the learners reanalyse their initial assessment? We chose to investigate L1 Spanish – L2 English advanced speakers for two reasons, the first one being availability since we are located in Spain, we have easy access to participants and the second one being the interchangeability of Spanish and Greek, because they are both what we call *null-subject* languages, we will explain what this term means in the introduction section. To research this phenomenon, we are going to be using the eye-tracking methodology with a visual world paradigm experiment based on the one done by Cunnings et al., (2017), the model of the eye tracker is the EyeLink portable duo by SR Research. The population of the experiment is $n = 12$, 9 males 3 females, all from the English studies degree at the University of Granada, who self-reported and averaged on the OQPT test a C1 level of English. The results seem to suggest that L1 Spanish – L2 English advanced speakers do acquire the English anaphora resolution strategy, it has also been seen that when presented with a contradicting noun L1 Spanish – L2 English learners seem to reanalyse correctly the sentence. Having said all of this, our results coincide with the most recent literature in the field and specifically with the paper by Cunnings et al., (2017) which is what we expected.

1. Introduction

In this bachelor's thesis we are going to try and shed light on the question of whether sentence processing in a second language is fundamentally similar or different to sentence processing in native language L1 (Cunnings et al., 2017).

To this aim we firstly read articles relevant to this area, anaphora resolution (AR) and second language acquisition (Alonso-Ovalle et al., 2002; Contemori et al., 2019; Conklin & Pellicer-Sanchez, 2016; Cunnings et al. 2017).

AR in Spanish can be processed in various ways, the first one is using a noun phrase (NP), the second one is an explicit pronoun (EP) and the third one with a null ($\emptyset$) pronoun as subject.

(1) Antonio fue al supermercado.

Anthony went to the supermarket.

(2) Él fue al supermercado.

He went to the supermarket.

(3) $\emptyset$ fue al supermercado.

$\emptyset$ went to the supermarket.

Above are examples of the realizations of AR in Spanish, (1) being the NP, (2) being the EP, and (3) being the $\emptyset$ pronoun. These different realizations are used to indicate information about the topic discourse, there are two types of topic movement, these being topic continuity (TC) and topic shift (TS).

(4) Juan_i discutía con Jorge_j por la comida mientras $\emptyset$ _{#i} limpiaba la mesa. (TC)

John argued with George for the food while $\emptyset$ cleaned the table.

(5) Juan_i discutía con Jorge_j por la comida mientras él_{#j} limpiaba la mesa. (TS)

John argued with George for the food while he cleaned the table.

Above are two examples of TS and TC in Spanish, the first one (4) being TC, this means that the $\emptyset$ pronoun is referring to Juan and not Jorge, therefore conserving the topic of discourse. The second one (5) being TS, this means that the EP refers to Jorge instead of Juan, therefore changing the topic of discourse.

AR in Greek can also be processed in various ways, with a $\emptyset$ pronoun and with an EP (aftós) (Cunnings et al., 2017)

(6) Kathós o Pétros_i kathótan me ton Jáni_j sto leoforío, $\emptyset$ _{#i} vríke mia tsánta káto apó to káthisma. (TC)

As Peter sat with John on the bus, $\emptyset$ found a bag under the seat.

(7) Kathós o Pétros_i kathótan me ton Jáni_j sto leoforío, aftós_{#j} vríke mia tsánta káto apó to káthisma. (TS)

As Peter sat with John on the bus, he found a bag under the seat.

Above are two examples of TS and TC in Greek, the first one (6) being TC, this means that the Ø pronoun is referring to Pétros and not Jani, therefore conserving the topic of discourse. The second one (7) being TS, this means that the EP refers to Jani instead of Pétros, therefore changing the topic of discourse.

As it can be seen in the examples given previously Spanish and Greek behave similarly in the topic shift and topic continuity aspect of AR.

In contrast, we have English, English is a non-null subject language which means that Ø pronouns are not allowed to be used which then makes sentences like (8) ungrammatical.

(8) John argued with George for the food while Ø cleaned the table.

This makes English a more ambiguous language since it is obligated to use overt pronouns at all times and therefore there is no way to mark antecedent preference as in Spanish or Greek. Resulting in what is called a subject preference strategy.

Research has been pursued to see whether speakers from non-null subject backgrounds acquire the preferences of null and overt pronouns from null-subject languages. Notwithstanding, less research has been carried out on whether L2 learners with a null-subject language background acquire the preferences from non-null subject languages in a native-like way. Studies like Cunnings et al. (2017) were pioneers at addressing this niche in the research of anaphora resolution and reanalysis.

In his original study Cunnings et al. (2017) performed two experiments that supported each other, the first one investigated whether the L1 Greek-L2 English speakers processed English in a native-like way. The second one addressed the question of whether the processing strategy in Greek was as thought, as stated in the examples (5) and (6). Being this a final dissertation for the English Studies degree the scope and size will not be able to comprise a replication of the two experiments, and therefore we will be replicating only the first one.

To this aim we report a visual world paradigm study investigating anaphora resolution of L1 Spanish – L2 English speakers, based on the methodology used by Cunnings et al. (2017).

2. Literature review

2.1. Previous Research on Anaphora Resolution

Anaphora resolution (AR) is the phenomenon where an element such as an NP, an EP or a Ø pronoun have a pragmatic connection to another element in a given sentence. This phenomenon, as everything in language, has some rules that decides how it works. The Preference of Antecedent Hypothesis (PAH) was one of the first suggestions to cover this niche in the research, postulated by Carminati (2002), the PAH suggested that there was a

division in the preferences of elements to process AR, for example it proposed the idea that a Ø pronoun was more likely to be used to refer to a subject antecedent than an object antecedent. To test the PAH Carminati (2002) used 40 Italian students from the University of Padua, with a series of online and offline experiments. For the experiments she used sentences such as (9) and (10).

(9) Quando Vanessa_i ha visitato Giovanna_j in ospedale, Ø_{#i} le ha portato un mazzo di fiori.

When Vanessa_i has visited Giovanna_j in hospital, Ø_{#i} to her has brought a bunch of flowers.

(10) Quando Vanessa_i ha visitato Giovanna_j in ospedale, lei_{#j} era già fuori pericolo.

When Vanessa_i has visited Giovanna_j in hospital, she_{#j} was already out of danger.

(Carminati, 2002)

The first sentence (9) with a Ø pronoun suggests a bias towards the subject following the PAH and the second sentence (10) with an EP suggests a bias towards the object also following the PAH. The overall results concluded that anaphora resolution in Italian is highly structure-dependent, and the Spec IP position plays a deeply important role in it. Later that same year Alonso-Ovalle et al. (2002) examined the PAH to investigate whether it was capable of predicting the preferences of elements to process AR in Spanish. It makes sense that they would want to test the PAH on Spanish since it shares similar features, being both null subject languages. Eighty students at the Universidad Complutense de Madrid participated in Alonso-Ovalle et al. (2002), they used sentences similar to (11) and (12).

(11) María ayudó a Lucía. Ø está agradecida.

Mary helped Lucy. Ø is grateful.

(12) María ayudó a Lucía. Ella está agradecida.

Mary helped Lucy. She is grateful.

After reading the sentence a question would be presented for them to choose a referent for the second sentence subject. The results demonstrated that when a Ø pronoun was the subject of the second sentence, as in (11), a 73.2% chose the subject of the first sentence as antecedent. In counterpoint, when a EP was the subject of the second sentence, as in (12), the percentage of selection towards the subject decreased to 50.2%. Now the question was do bilinguals work in this same way we have seen monolinguals work?

To address this question different experiments have been performed, Cunnings et al., (2017) reported two visual world paradigms experiments with L1 Greek monolinguals, and L1 Greek – L2 English bilinguals, the two experiments supported each other. They used audio and image processing to tackle the AR processing question, they constructed sentences similar to (13), (14), (15) and (16).

- (13) Mr. Smith_i found Peter_j by the path. Immediately afterwards he_{#j} drank a couple of glasses of water. (object bias ambiguous)
- (14) Mrs. Jones_i found Peter_j by the path. Immediately afterwards he_{#j} drank a couple of glasses of water. (object bias unambiguous)
- (15) Peter_i found Mr. Smith_j by the path. Immediately afterwards he_{#i} drank a couple of glasses of water. (subject bias ambiguous)
- (16) Peter_i found Mrs. Jones_j by the path. Immediately afterwards he_{#i} drank a couple of glasses of water. (subject bias unambiguous)

In the sentences above we can see 4 types of different structures, the first one presents two characters, both male, and a scenery. Subsequently an overt pronoun appears which should set off the PAH and make the participant fixate on the picture of the Mr. Smith instead of Peter's, the second part of the sentence is the desambiguating noun which triggers the reanalysis since the "glasses of water" will always be held by Peter. Later on the literature review we will discuss the reanalysis observations done by Cunnings et al., (2017).

The novelty of their study was that they introduced a methodology that was barely being used at the time, eye-tracking. Eye-tracking is a technique that allows the linguist to collect information from the unconscious part of the brain that controls the impulses and preferences we might have, using this technique implies certain assumptions which we will discuss later on the methodology section. Cunnings et al., (2017) found that there is a preference for the subject antecedent in the English language when overt pronouns are present. Additionally, Contemori et al., (2019), did 6 experiments to also respond to the open questions of whether L2 speakers can reach nativelike competence on AR and how discourse complexity may have impacted previous results in other research. The experiment that pertains to this literature review is the second one. A total of 24 native English speakers and 31 intermediate proficient learners of English participated in this experiment. The materials for this second experiment were translations of Spanish sentences they used in their first experiment, (17) and (18) are examples of the structures they used for the experiment.

- (17) Julia saludó a María mientras que ella estaba en el supermercado.

Julia greeted Mary while she was in the supermarket.

- (18) Mientras que ella estaba en el supermercado, Julia saludó a María.

While she was in the supermarket, Julia greeted Mary.

After each sentence a multiple-choice question (19) was presented to them to collect the information of how the AR was processed.

- (19) ¿Quién estaba en el supermercado?

Who was in the supermarket?

- a) Julia
- b) Maria

- c) Otra persona
Someone else

The results suggested that the L1 Spanish – L2 English speakers did acquire the PAH, now PAS (Position of Antecedent Strategy), of the English language. Ultimately it was found that they interpreted AR in an English nativelike manner, associating the overt pronoun to the subject antecedent. They remark how these results coincide with the results of Cunnings et al., (2017) and Contemori & Dussias (2015).

Finally to end this part of the literature review, we would like to mention how this behaviour, the PAS, has not only been seen in experimental settings forcing answers out of the participants through questions. Lozano & Quesada (2020) found similar PAS behaviour in corpora data, the preference of overt pronouns to refer to subject antecedents, in addition to this they also encountered that L2 speakers preferred to be overexplicit or redundant instead of ambiguous in their utterances.

2.2.Reanalysis during L2 processing

Ambiguation is a tool used by many linguists to see preferences in analyses by native and bilinguals. Reanalysis is the process when a noun forces an analysis on a sentence which already had been analysed and assessed by the reader. The first study to address this phenomenon was Dussias & Cramer Scaltz (2008) who found that participants decelerated reading speeds while processing in the L2. The participants were 59 L1 English monolinguals as a control group, and 60 L1 Spanish – L2 English bilinguals. They were tested using a self-paced reading task (SPRT), the stimuli were presented in a non-cumulative reading moving window, after reaching the end of each sentence, they were asked yes or no questions which they answered by pressing two different buttons. They recorded the reaction times for these questions, and they found out that the deceleration in reading times could be due to attempted reanalysis, nonetheless this cue does not address if the reanalysis was finished correctly or not. To this issue some studies have investigated whether reanalysis in a nativelike way is in fact possible, technically it should be possible because if anaphora resolution in L2 has been observed to behave in a nativelike way (Cunnings et al., 2017; Contemori et al., 2019; etc.) reanalysis should behave in a similar way, since it is a key part of language processing. Two studies, the first one Roberts & Felser (2011) reported on this question, they tested advanced L2 learners versus L1 speakers for garden-path sentences such as (20)

(20) “While the band played the song pleased all the customers.” (Roberts & Felser, 2011).

After reading the sentence for meaning they were presented with a question they had to answer, as in (21).

(21) “Did the song please the customers?” (Roberts & Felser, 2011)

Specifically, they tested 24 L1 Greek – L2 English speakers, and a control group of 24 native English speakers. They all were students from the University of Essex. It was observed that L2 learners were less accurate than L1 speakers on the comprehension questions. Nevertheless, in contrast, the second study done by Hopp (2015) found similar quite high rates of accuracy in comprehension questions between L1 speakers and L2 learners, which suggests the idea of reanalysis in a nativelike manner is possible. From this conclusion a question that arises is whether if the initial analysis after reanalysis completely disappears or does it remain in the language domain (Slattery et al., 2013). To this end, Jacob and Felser (2016) investigated this phenomenon using an eye-tracking experiment that used sentences like (22)

(22) “While the gentleman was eating the burgers were still being reheated in the microwave.

Question: Was the gentleman eating the burgers?” (Jacob & Felser, 2016)

They found decelerations in reading times after the disambiguating part of the sentence, suggesting a possible attempt to reanalyse the sentence. In addition, L2 learners made more errors when answering the trial questions suggesting increased rigidity in an initial analysis compared to the L1 speakers. Finally, we have the reanalysis presented by Cunnings et al. (2017) using a biasing noun phrase which disambiguates the sentences where the two antecedents have the same gender, as in (23), (24), (25), (26). The reader cannot resolve the ambiguity using gender information and must use the PAS as an unconscious guide. As the biasing NP appears, in this case “a couple of glasses of water”¹, it triggers the reanalysis in the reader, making him fixate on Peter rather than Mr. Smith.

(23) Mr. Smith_i found Peter_j by the path. Immediately afterwards he_{#j} drank a couple of glasses of water. (object bias ambiguous)

(24) Mrs. Jones_i found Peter_j by the path. Immediately afterwards he_{#j} drank a couple of glasses of water. (object bias unambiguous)

(25) Peter_i found Mr. Smith_j by the path. Immediately afterwards he_{#i} drank a couple of glasses of water. (subject bias ambiguous)

(26) Peter_i found Mrs. Jones_j by the path. Immediately afterwards he_{#i} drank a couple of glasses of water. (subject bias unambiguous)

It was observed by Cunnings et al. (2017) that in the object bias unambiguous structure the subject antecedent was almost never selected, while in the subject bias unambiguous the subject antecedent was almost always selected. Curiously enough, he found that the ambiguous conditions were linked almost always to the subject antecedent, meaning that the PAS was probably not overwritten.

¹The biasing noun phrase appeared always on the hand of the target antecedent (i.e.: Peter) so to resolve the ambiguity and force the reanalysis.

3. Research questions

Considering the body of literature on AR that has been reviewed, we hereby propose the following research investigation.

RQ1. Do L1 Spanish – L2 English speakers acquire nativelike processing of English AR?

H1. L1 Spanish – L2 English speakers will acquire nativelike processing of English AR.

Due to the literature we have seen in AR during L2 processing (Cunnings et al., 2017; Contemori et al., 2019; Lozano & Quesada, 2020; etc), we predict that L1 Spanish – L2 English advanced speakers will be able to unconsciously adopt the PAS of the English language without major problems.

RQ2. Do L1 Spanish – L2 English speakers reanalyse a sentence correctly if presented with a biasing noun?

H2. L1 Spanish – L2 English speakers will reanalyse a sentence correctly when presented with a biasing noun phrase? We expect to see similar reanalyses patterns as the ones seen by Cunnings et al., (2017), due to the similarities in the objectives and design of both experiments.

RQ3. Does the level of proficiency in the L2 affect the rigidity of the reanalyses ?

H3. The most advanced L1 Spanish – L2 English speakers will be less rigid in their reanalyses. As seen in relevant literature mentioned before, Hopp (2015), Cunnings et al., (2017), etc; we expect to see that the participants who are more proficient in their L2 to be less rigid in their reanalysis.

4. Methodology

The methodology used for this experiment is Eye Tracking as Cunnings et al. indicates in his experiment. Eye tracking is the process by which eye movements are measured – typically in order to determine where (or what) someone is looking at – also known as the “point of gaze”. Examples of uses for research are varied from psychologists and linguists studying the processes of reading in the eye movements to other researchers studying the eye movements when judging emotions on facial expressions. In addition to tracking gaze, modern eye trackers can provide other useful measures such as pupil size (for pupillometry research) and blink rate (SR Research, 2023).

One of the biggest technological changes over the last couple of decades has been the near-universal adoption of video-based eye tracking as the technique of choice. So how do video-based eye trackers actually work?

At the heart of all video-based eye tracking is a camera (or cameras) that takes a series of images of the eye. The EyeLink Portable Duo uses a single camera that is capable of taking up to two thousand images of both eyes every second. Within three ms from the image of the

eye being taken, EyeLink systems work out where on the screen the participant is looking and relay this information back to the computer controlling stimulus presentation. So how is this done? The eye-tracking software uses image processing algorithms to identify two key locations on each of the images sent by the eye-tracking camera – the centre of the pupil, and the centre of the corneal reflection. The corneal reflection is simply the reflection of a fixed light source (the infrared illuminator) that sits next to the camera (SR Research, 2023).

The visual world paradigm which is the specific model of experiment that we will be using in this thesis (Cunnings et al., 2017; Jacob & Felser, 2016; etc) is very useful since it provides with the time-course of processing spoken language comprehension. On a typical visual world experiment the participants will firstly be provided with some instructions. Then as the experiment starts, they will see two or more images, in ours three, and hear some sentences, after this they might hear an instruction or a “ding” which instructs them to select an image of the ones they were seeing on the screen. An example of a trial of a visual world paradigm might be figure 1.

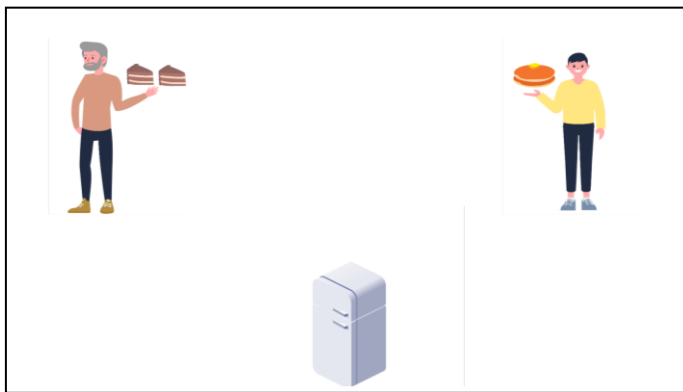


Figure 1. A trial of a visual world paradigm experiment.

4.1. Participants

Gender	Frequency	Percentage
Male	9	75.0
female	3	25.0
English level	Scores OQPT	Corresponding level
max. score	59	C2
min. score	48	C1
avg. Score	52	C1
Age		
max. Age	23	
min. Age	19	
avg. Age	21	

Figure 2. Table of participants

They were L2 Spanish natives (9 males and 3 females), from academic background, specifically from the English studies degree at the University of Granada, which self-reported a C1 level of proficiency in the English Language on average, additionally they were assessed using the OQPT (Oxford Quick Placement Test) where on average the participants achieved 52 out of 60 points which corresponds with the self-reported level. All of the participants were tested in the Lab space “BilinguaLab” at the University of Granada. As for the English native control group we are taking as reference the one in the Cunnings et al. (2017) paper, we will discuss the reasons behind this decision in the limitations section of this paper. All L2 Speakers learned English from the age of five or later in school, used English for academic purposes and enrolled into degree programs in which the curriculum is taught in English as mentioned above. They received a ticket for one coffee at the vending machine in the faculty for their participation.

4.2. Eye-tracking measures

The measures that the eye-tracker is capable of recording the terms that are usually used in this field.

Fixations:

A pause in eye movement in which the retina is stabilized over a stationary object. It is estimated that 90% of viewing time is comprised of fixations, which can last between 150-600 ms. (Cullipher et al., 2018)

Saccade:

A rapid movement of the eye that occurs between fixations. During this transition, the viewer is essentially blind. Saccadic movements can be both voluntary and reflexive, so careful consideration is important in analysing data related to saccades. (Cullipher et al., 2018)

Area of interest (AOI):

A region of the visual stimulus in which measurements, primarily fixations, can be aggregated as part of analysis. AOIs can be defined in two ways. First, AOIs can be defined by the researcher before data collection, a technique that is commonly used when the researcher wants to know if a participant looks at a particular region of the visual stimulus. In the second method, AOIs are defined after data collection based on cluster analysis. Cluster analysis aggregates fixation data from all participants to indicate regions of the visual stimulus that have a high concentration of fixations. (Cullipher et al., 2018)

4.3. Software

The software used has been Excel, which is a spreadsheet developed by Microsoft and that allowed us to organize and design the written stimuli. Audacity, which is a free and

open-source digital audio editor, it permitted us to record said stimuli. Inkscape is another free and open-source program, it enabled us to design and modify copyright-free vector graphics; this was highly useful since they have qualities such as resizability without losses in quality or lightweight source files. Also Experiment Builder was used to implement all the stimuli in a framework where rt and eye-movement are recorded as well as Data Viewer to analyse the obtained data.

4.4. Hardware

The hardware used throughout the experiment has been the eye tracker “EyeLink portable duo” by SR Research at a sampling rate of 500Hz on the right eye, with the chin rest mounted, a computer with Windows 10, and a wired mouse to answer the comprehension questions.

To record the stimuli another computer with Windows 10, a professional cardioid microphone, and Audacity were used.

The person who read the sentences was an English Native with an intended standard accent.

4.5. Design

Experimental and filler items were pseudo-randomised with a Latin-square design. Additionally, three practice items appeared at the start of each list giving the participant the possibility to practice the process that experiment followed. Ahead we can see a design table example we used for the experiment to design each list and the Latin square design. In the appendices we have included all the sentences following the style shown in figure 4.

Item	Sentences	Lists
1	Object Bias Ambiguous	1
1	Subject Bias Ambiguous	2
1	Object Bias Unambiguous	3
1	Subject Bias Unambiguous	4
2	Object Bias Ambiguous	2
2	Subject Bias Ambiguous	3
2	Object Bias Unambiguous	4
2	Subject Bias Unambiguous	1
3	Object Bias Ambiguous	3
3	Subject Bias Ambiguous	4
3	Object Bias Unambiguous	1
3	Subject Bias Unambiguous	2
4	Object Bias Ambiguous	4
4	Subject Bias Ambiguous	1
4	Object Bias Unambiguous	2
4	Subject Bias Unambiguous	3

Figure 4. Design example with Latin square design.

4.6. Stimuli

As seen above the sentences followed this pattern: S + Vb + Ob + C. Adv + S + Vb + Adv + Np. An example sample of the sentences are (27), (28), (29), and (30).

(27) Mr. Smith scolded Peter by the fridge. Immediately afterwards he ate a couple of pancakes.

(28) Mrs. Jones scolded Peter by the fridge. Immediately afterwards he ate a couple of pancakes.

(29) Peter scolded Mr. Smith by the fridge. Immediately afterwards he ate a couple of pancakes.

(30) Peter scolded Mrs. Jones by the fridge. Immediately afterwards he ate a couple of pancakes.

There were 16 experimental items of each condition. To accompany these experimental items, we devised 24 fillers and distractors, these would be constant throughout the different lists in the experiment, they were designed to mix the structure and relations between the referents and scenarios. Some examples of the fillers and distractors are (31), (32), and (33).

(31) Before Peter talked to Susan by the kitchen, the kitchen was quite dirty.

(32) The wheelbarrow was very heavy, but Mrs. Jones was helping Peter.

(33) The lawyer was talking to Mrs. Jones by the building, the building was huge.

Also, for practice we created 3 practice items which would as well remain constant throughout the experiment. In addition, a brief interview was done previously to the experiment to help relax the participants and activate the second language.

4.7. Instructions

The first thing the participants saw in the trial was an instruction saying that before anything else, a calibration was needed (figure 5). We used the autocalibration option that the EyeLink laptop provides, after the calibration is properly done a validation is needed. The validation seemingly repeats the same process as the calibration in order to check that the calibration is in fact correct.

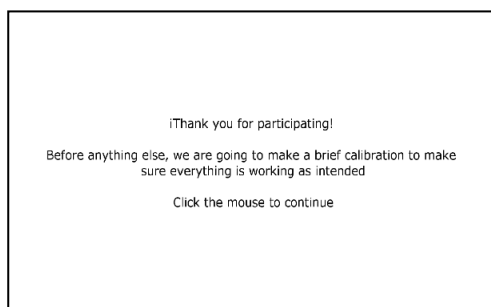


Figure 5. Greeting message.

Secondly after the calibration had been done, the experimental instructions appeared on the screen, they explained the flow of the experiment. They can be seen in Figure 6.

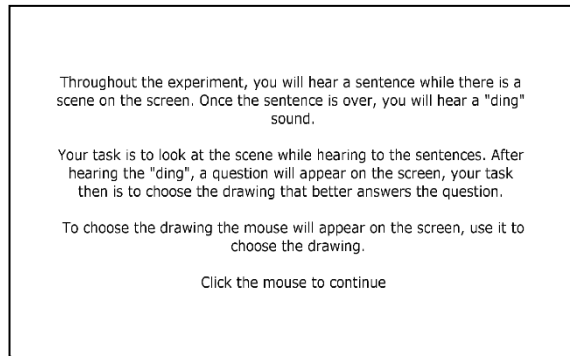


Figure 6. Instructions for the process of the experiment.

Thirdly and before the practice we presented the key characters in the sentences so they could familiarise themselves with them before the experiment. In addition to these three characters there were also 5 generic characters (a lawyer, a waiter, a plumber, a doctor, and a young girl “Susan”, these can be seen in figure).

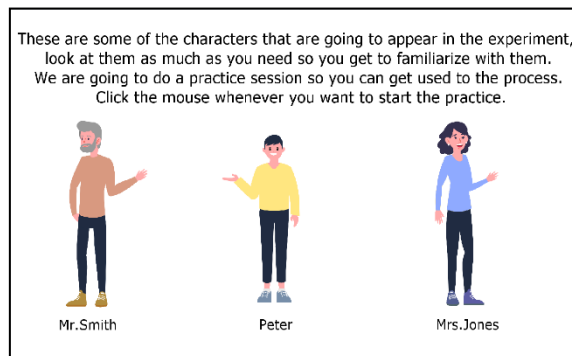


Figure 7. Characters presentation.

The practice sentences were designed specifically for practice therefore they included (34) a unique experimental item, (35) and (36) which were distractors.

(34) Mr. Smith kissed Mrs. Jones in the garden. Immediately afterwards he presented a beautiful ring.

(35) Peter informed the chef in the kitchen, the kitchen was a mess.

(36) The lawyer was reimbursing the waiter for the stolen box. The waiter looked weary.

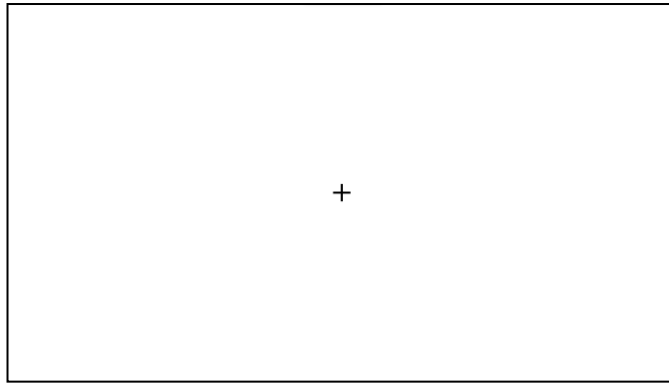


Figure 8. Fixation Cross.

The figure 8, is what they saw before a specific trial, we call it a Fixation Cross. We use the fixation cross to fix the gaze at the centre of the screen so there is no bias before the next trial.

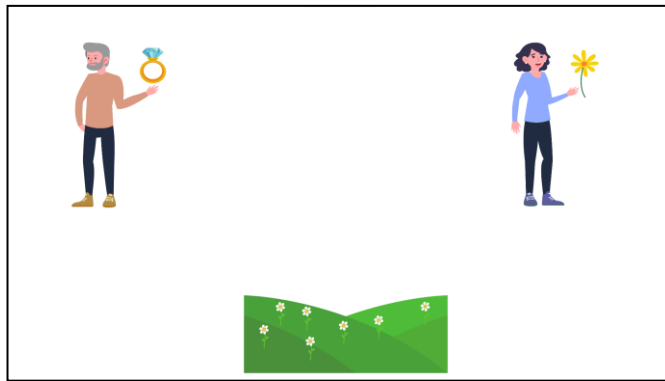


Figure 9. Characters appears with objects while sentence is playing.

The figure 9 is an example of how the pictures are design to appear on the screen, the characters would be shown on the sides of the screen with their objects (one of them being the biasing NP) and the scenery at the centre bottom.

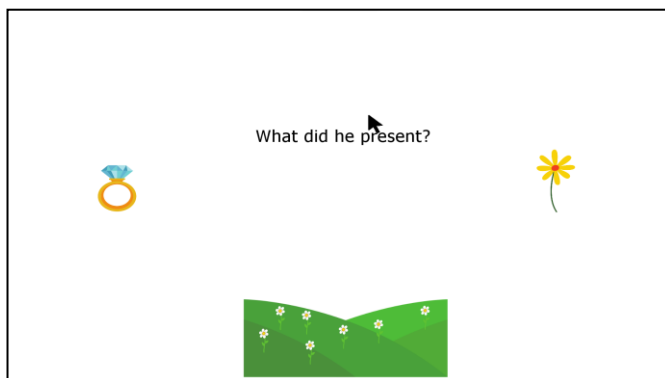


Figure 10. Objects remains.

In figure 10 the comprehension question emerges, this specific part of this specific item was designed to give an insight on how the experiment worked but the comprehension question was not experimentally interesting, and therefore it was designed in a distractor way to not give an exact clue of what the experiment was about. They used a wired mouse to answer the question by selecting the image they saw fit.

4.8.Data analysis

The data analysis is going to be performed in the software called “Data Viewer”, which is an analytical tool, it allows the user to the display, filtering, and report output of several EyeLink eye-trackers, including ours the EyeLink portable duo.

4.8.1.Heatmaps

A heatmap is a specialized method where data values are graphically represented with colours, in data viewer we have this method, and it can be applied to the eye gaze fixations creating quite interesting images from which we can infer and interpret results.

To go in a bit of depth of how the heatmaps work, or how SR Research calls them “fixation maps”. Basically to create a fixation map what data viewer does is create a “2D Gaussian” on each of the fixations – the Gaussian centre is the fixation location, the width of the Gaussian is influenced by an adjustable sigma value in degrees of visual angle (the larger the sigma value, the larger the area that will be influenced by the fixation), and the height of the Gaussian can be weighted either equally for all fixations, the internal map is normalized so it can be applied to the colour or brightness scale used to create an aggregate view.

4.8.2.Interest Periods

Besides the heatmaps definitions, the user may also selectively view and perform analysis on data within a specific period of time within a trial (Interest Period) which is needed to see the time windows that we mention ahead in the paper. For example, the user may want to examine the eye data between two important markup messages; for a trial with multi-page manipulation, analysis may be performed separately for the period of each page presentation. The EyeLink Data viewer allows the user to create interest periods based on messages, button events, or duration, as seen in figure 11. To do that, when programming an experiment, the user should mark up possible interest periods with messages or other important events such as button presses. If different images were shown within different interest period, the user should also make sure that messages used to load images, text, or draw simple graphics be recorded with the scope of the intended interest period. (SR Research, 2023)

Figure 11. Design of an interest period in Data viewer.

4.9. The pronoun time window

As we mentioned we controlled for the time window for the pronoun to be on average about 1100ms. Experiment Builder allows the user to set what is called a “timer” and what these “timers” (i.e.: `TIMER_PRONOUN` in figure 7) serve in our experiment in contrast to Cunnings et al. is that Experiment Builder will let you set them in order to be able to generate the heatmaps we are going to be using in this paper.

4.10. The biasing noun time window

We also controlled for the time window for the biasing noun to be on average of 900ms. As it can be seen in figure 12, we had another timer for the biasing noun which had the average time of onset for the noun on it so we could generate a heatmap after the onset of the noun.

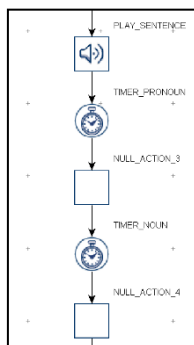


Figure 12. Experiment structure in Experiment Builder.

4.11. Comprehension questions

500ms were left after the sentence ended, subsequently a “ding” sound appeared indicating the end of the sentence for the participant, another 500ms were left. Immediately afterwards,

the comprehension question was displayed, following Cunnings et al. fashion. An example of the comprehension question with its sentence would be (37).

(37) Mr. Smith hugged Peter by the tree. Immediately afterwards he picked a couple of green leaves.

Question: Who picked the leaves?

5.Results

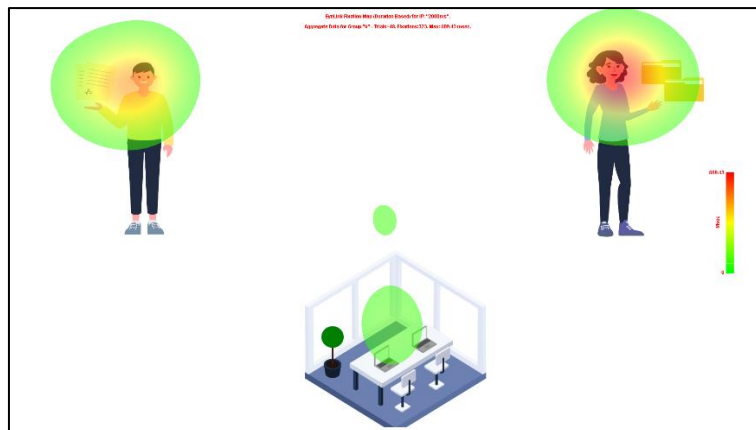
For the results we obtained the following heatmaps, we will discuss them in the discussion section ahead in the paper.

5.1.Heatmaps.

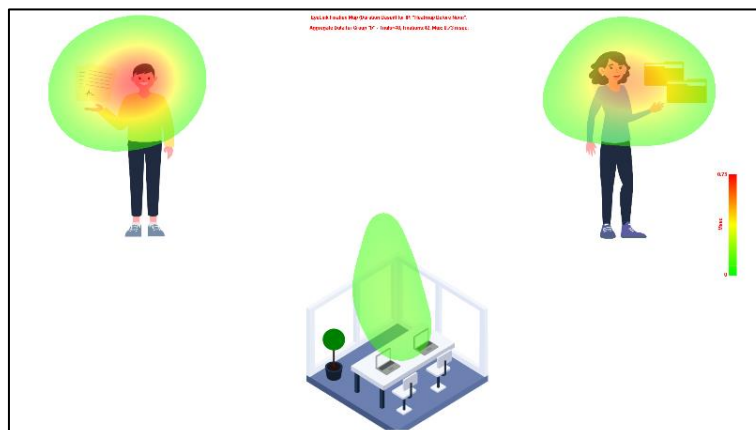
5.1.1. OB UN (Object bias unambiguous)

e.g. : **Mrs. Jones** helped **Peter** at the office. Immediately afterwards **he** sent couple of documents.

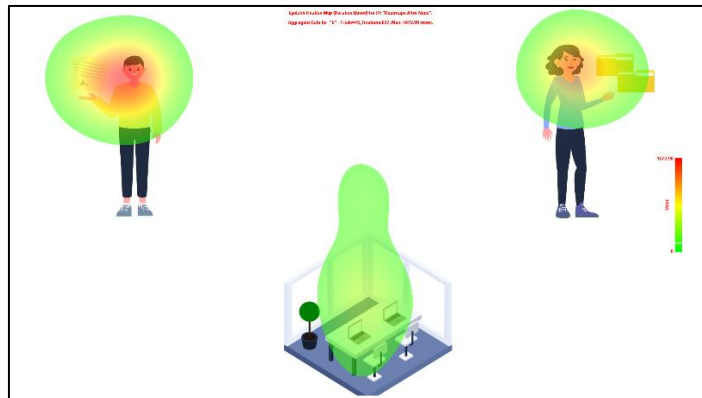
After hearing **Mrs. Jones** and **Peter** we took this heatmap, which clearly marks a preference for **Mrs. Jones**.



After hearing the pronoun **he** there is a clear shift in preference towards **Peter** as it matches gender.



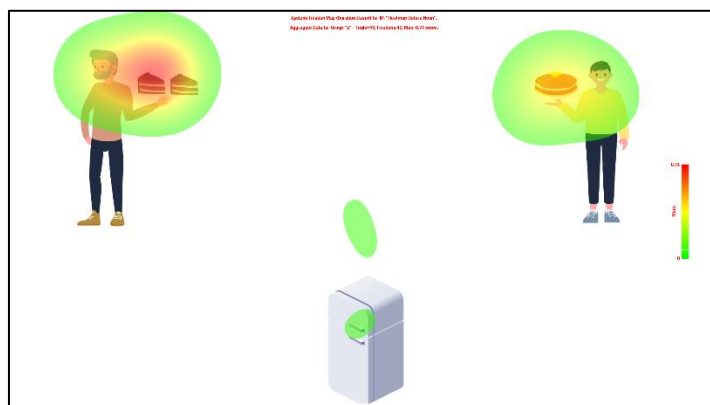
In this heatmap the biasing noun phrase, in this case “**a couple of documents**”, has appeared and the preferences does not seem to shift much.



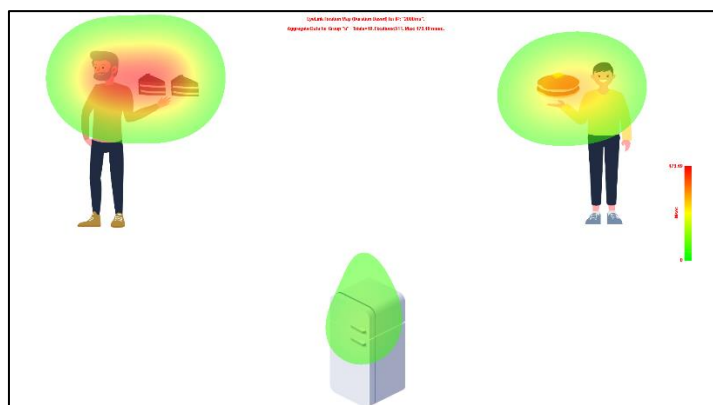
5.1.2. OB AM (Object bias ambiguous)

e.g. : **Mr. Smith** scolded **Peter** by the fridge. Immediately afterwards **he** ate a couple of pancakes.

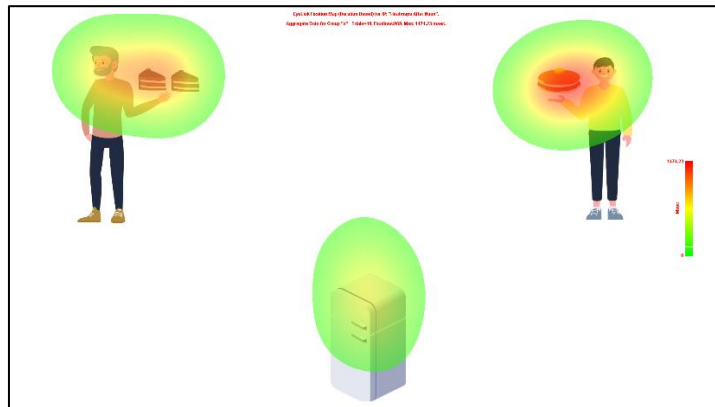
In this condition, firstly we see a bias towards **Mr. Smith**. Only the two antecedents have been heard by the participant.



After hearing the pronoun **he** the preference stays some what the same, and even it could be said that increases towards **Mr. Smith**.



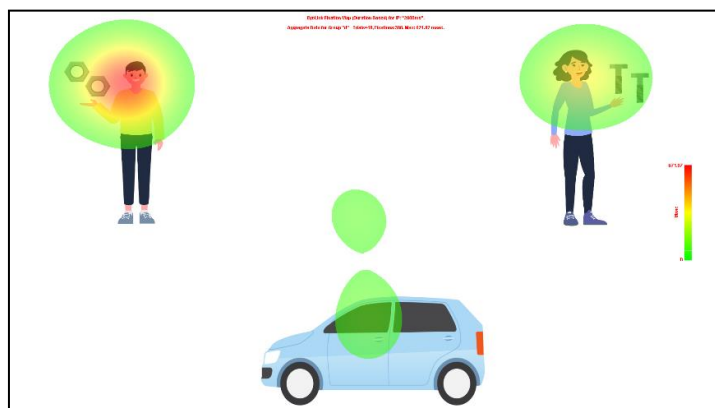
Finally after hearing the biasing noun phrase, in this case “**a couple of pancakes**”, there is a shift towards **Peter**.



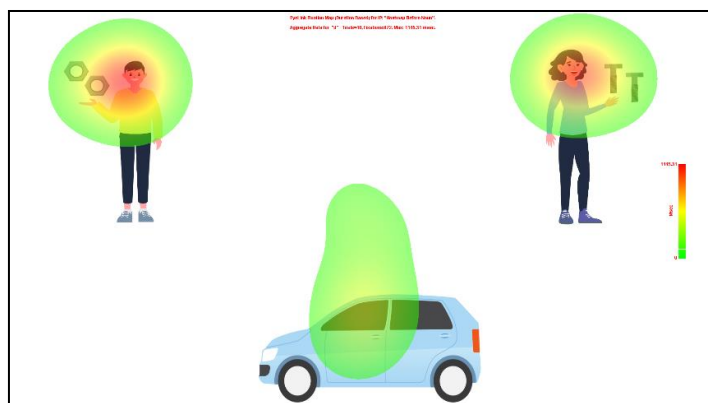
5.1.3. SB UN (Subject bias unambiguous)

e.g.: **Peter thanked Mrs. Jones by the car. Immediately afterwards he held a couple of metal nuts.**

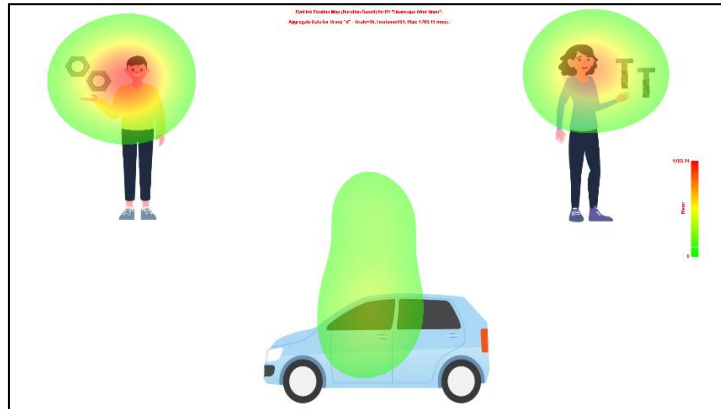
In this third condition, firstly we see a preference for the subject **Peter**.



Secondly, the pronoun **he** appears and there seems to be a bit of a shift towards **Mrs. Jones**.



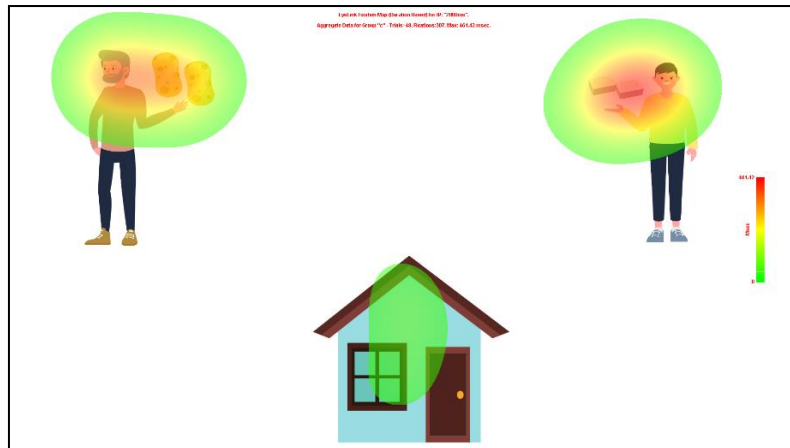
Finally after hearing the biasing noun phrase, in this case “**a couple of metal nuts**”, the participants seem to reaffirm their initial analysis.



5.1.4. SB AM (Subject Bias Ambiguous)

e.g.: **Peter** helped **Mr. Smith** by the house. Immediately afterwards **he** bought a couple of soap bars.

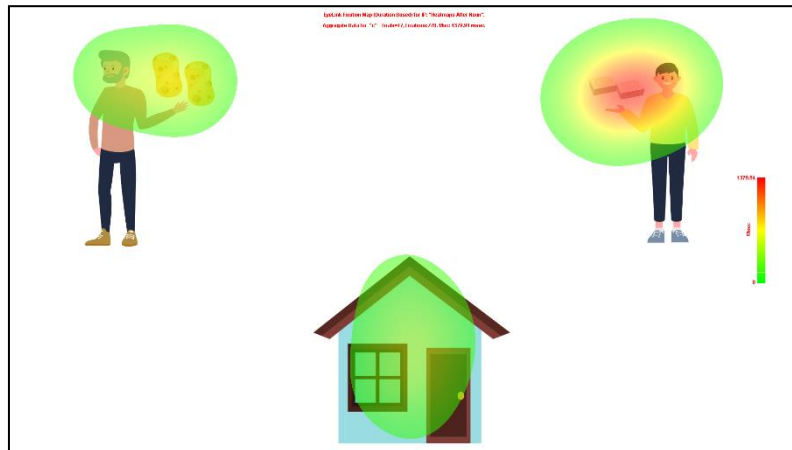
Initially after hearing both of the antecedents, **Mr. Smith** and **Peter**. It seems there is in fact a preference towards **Peter**, the subject antecedent.



After hearing the pronoun “**he**” there seem to be a confirmation on the initial analysis.



Finally, the initial analysis is confirmed by the biasing noun phrase, in this case “a couple of soap bars”.



6. Discussion

Now we are going to go through each condition and discuss every heatmap in detail.

6.1. First Condition OB UN: Mrs. Jones verb Peter complement phrase. Adverb phrase he verb noun phrase.

We can see that after the onset of the object, in this case **Peter**, the participant fixated more on the subject **Mrs. Jones** rather than on the object **Peter**, which is surprising because it means that the subject preference strategy starts to apply even before the pronoun is even mentioned. In the second image we can see how after hearing the pronoun **he** there is a clear shift in preference towards the object because the gender matches, which corresponds to the shift in the colours of the images. As the sentence ends with the **noun phrase**, we can see how the size of the area highlighted for **Mrs. Jones** downsizes even more compared to the image previous, but nonetheless the preference for the subject makes some fixations to remain on **Mr. Jones**.

6.2. Second Condition OB AM: Mr. Smith verb Peter complement phrase. Adverb phrase he verb noun phrase.

We can see a somewhat different pattern, firstly we can see there is a clear preference even before hearing the pronoun for the subject **Mr. Smith** as it happened with the first condition, after hearing the pronoun **he** the first analysis remains the same and even the size of the red area is even bigger, but then the biasing **noun phrase** appears and forces a reanalysis shifting the red area completely towards **Peter**. It is remarkably interesting that still some colour remains on **Mr. Smith** because that suggests that the subject preference strategy has a strong influence even if a reanalysis has gone through.

6.3. Third Condition SU UN: Peter verb Mrs. Jones complement phrase. Adverb phrase he verb noun phrase.

We can see perfectly how there is firstly a clear preference for the subject **Peter** over the object **Mrs. Jones**, even before the pronoun **he** appears. Somewhat interestingly when the pronoun **he** is heard by the participant there seems to be a bit of doubt on the preference although it is still very clear that it is **Peter** the preferred antecedent because the colour is more intense. Finally, the **noun phrase** pops up in the audio and confirms the analysis on **Peter** being the preferred antecedent.

6.4.Fourth Condition SU AM: *Peter verb Mr. Smith complement phrase. Adverb phrase he verb noun phrase*

We can also see a very clear and good preference for firstly **Peter** but there are also some fixations for **Mr. Smith**, and what is really interesting comes when the pronoun **he** is heard by the participant, because we see that the preference for the subject is very present, but the ambiguity remains; therefore **Mr. Smith** is a bit illuminated. And at the end the biasing **noun phrase** appears resolving the ambiguity and forcing the reanalysis of the participants that were still doubting.

These results suggest differences between Spanish speakers resolving overt subject pronouns in their L1 compared to Spanish speakers resolving overt subject pronouns in L2 English.

7.Conclusion

The results of experiment indicate that L1 Spanish speakers of L2 English process and interpret overt pronouns in a largely “native-like” way. L2 English used gender information to select and resolve ambiguous pronoun as referring to the sentence subject and current TC. Cunnings et al. (2017) mentions that they were not able to find evidence of L1 Greek – L2 English processing overt pronouns in English as indexing TS. Similarly, we have found that with L1 Spanish - L2 English we have not found any clear evidence of speakers interpreting overt pronouns as indexing a topic shift, as in their L1 Spanish. This also corresponds with the study aforementioned by Lozano & Quesada (2020) who found in corpus production that L1 Spanish – L2 English tend to produce overt pronouns in the L2 to refer to TC instead of TS. According to Cunnings et al. (2017), the obtained results stand in contrast to the findings of Roberts et al., (2008), who observed that L1 Turkish – L2 Dutch speakers interpreted overt pronouns in Dutch as indicating TS, similar to their interpretation in Turkish. Cunnings et al. (2017) proposes that one potential reason for this disparity could be attributed to the varying language proficiencies of the L2 speakers involved. They suggest that the L2 speakers in their study might have possessed a higher level of proficiency in English compared to the Dutch proficiency of the L2 speakers examined in Roberts et al., (2008). Cunnings et al. (2017) in addition mentions that the range of scores in the proficiency tests are similar and that even though the test are not fully comparable, the similar ranges do not support the idea of the learners tested were systematically higher in proficiency. This current study being a final dissertation of an undergraduate degree took the self-reported

level and the OQPT as a truthful source of information and that the scores in the proficiency level tests of Cunnings et al. (2017) correspond to an average level of C1.

Additionally, it was found that the comprehension questions were almost always in the SU UN condition interpreted as subject referring, almost never as subject referring in the OB UN condition, in the SU AM and in the OB AM the pronoun was interpreted as referring to the sentence subject almost always, even if the display information biased against it. We, therefore, can say that our results coincide with those of Cunnings et al. (2017). To sum up throughout this experiment we researched anaphora resolution during L2 processing, following the structure and design from Cunnings et al. (2017) paper. Our results, which coincided with Cunnings et al.'s, indicate that L1 Spanish – L2 English speakers use gender information to guide pronoun resolution. L2 English speakers were able to process and interpret ambiguous pronouns in English in a largely nativelike way, even though their L1 display a different anaphoric resolution strategy. We also found, similarly as Cunnings et al. did that L2 speakers were unlikely to revise an initial interpretation compared to Cunnings et al. (2017) control group L1 English speakers. Together these results indicate that L1 and L2 English share processing preferences in the field of AR, although initial analysis are more rigid during L2 processing (Cunnings, Fotiadou, & Tsimpli, 2017).

8.Limitations

The main limitation of this study is the use of heatmaps, which are helpful for an initial analysis of eye-tracking data but have inherent restrictions in accurately assessing certain fixations. Additionally, the relevance of the obtained data in mathematical terms remains uncertain. Another challenge we encountered was the laborious and essential task of designing the images, ensuring consistency in art style, colour salience, size, and resolution. Time constraints also played a significant role, as more time would have allowed for deeper analysis and a more technical utilization of the Data viewer tool. Lastly, it is important to note that the tested population size was not sufficient for a truly representative study; for more rigorous research, a larger sample would have been necessary.

9.Further Research

There is potential for varied and undoubtedly fruitful further research in this niche in the near future, there are several examples of improvements and potential enhancements that could be implemented for this experiment. First and foremost, a crucial step would involve analysing the data in terms of fixations and probability. Unfortunately, due to time constraints and the significant challenges associated with this task, it was not feasible to undertake such analysis within the scope of this project. Heatmaps are a great tool for having an overview of the ideas that we want to analyse but cannot in any way replace probabilistic analysis. In addition, in the long term another branching from this research would be to see if neural networks which work with

anaphora resolution i.e.: “transformer models” like ChatGPT or similar do adopt this kind of knowledge and if so, how do they apply it in their conversations and how are they acquired by these complex mathematical algorithms. I believe there is still much more research left to be done in the fields of anaphora resolution and L2 reanalysis, hopefully we will be able to revisit this undergraduate final dissertation in the future.

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APPENDICES

The full design of the sentences for the experiment.

Antecedents	Sentences	Conditions	List	Latin Square
SU UN	Peter hugged Mrs. Jones by the tree. Immediately afterwards he picked a couple of green leaves.	d	1	4
OB AM	Mr. Smith helped Peter by the table. Immediately afterwards he ate a couple of nice steaks.	a	1	3
OB UN	Mrs. Jones thanked Peter by the car. Immediately afterwards he held a couple of metal nuts.	b	1	2
SU AM	Peter brought Mr. Smith to the museum. Immediately afterwards he bought a couple of postcards.	c	1	1
SU UN	Peter helped Mrs. Jones by the door. Immediately afterwards he bought a couple of oil cans.	d	1	1
OB AM	Mr. Smith helped Peter by the house. Immediately afterwards he bought a couple of soap bars.	a	1	4
OB UN	Mrs. Jones hugged Peter by the stove. Immediately afterwards he ate a couple of sandwiches.	b	1	3
SU AM	Peter scolded Mr. Smith by the fridge. Immediately afterwards he ate a couple of pancakes.	c	1	2
SU UN	Peter helped Mrs. Jones in the office. Immediately afterwards he sent a couple of documents.	d	1	2
OB AM	Mr. Smith found Peter by the path. Immediately afterwards he drank a couple of glasses of water.	a	1	1
OB UN	Mrs. Jones advised Peter by the tent. Immediately afterwards he bought a couple of oranges.	b	1	4
SU AM	Peter helped Mr. Smith by the shop. Immediately afterwards he chose a couple of rugby balls.	c	1	3
SU UN	Peter scolded Mrs. Jones by the sink. Immediately afterwards he cleaned a couple of forks.	d	1	3
OB AM	Mr. Smith found Peter at the bus stop. Immediately afterwards he bought a couple of tickets.	a	1	2
OB UN	Mrs. Jones thanked Peter by the factory. Immediately afterwards he fixed a couple of lightbulbs.	b	1	1
SU AM	Peter hugged Mr. Smith by the garden. Immediately afterwards he picked a couple of flowers.	c	1	4
Filler	Before Peter talked to Susan by the kitchen, the kitchen was quite dirty.	filler	1	4
Filler	The doctor saw the waiter working in the restaurant, it was a nice place.	filler	1	3
Filler	The restaurant was closing but Susan was happily talking to Mr. Smith.	filler	1	2
Filler	Peter was talking to the doctor by the fountain, the fountain was pretty.	filler	1	1
Filler	The waiter was serving the food to Mrs. Jones by the garden, the garden was nice.	filler	1	1
Filler	Mr. Smith was paying the plumber for fixing the pipes. The plumber was tired.	filler	1	4
Filler	After Susan helped Mr. Smith by the shop, the shop was quite organized.	filler	1	3
Filler	The plumber helped the waiter carry the wheelbarrow, it was quite heavy.	filler	1	2
Filler	The wheelbarrow was very heavy but Mrs. Jones was helping Peter .	filler	1	2
Filler	Susan was asking the waiter for some ice cold water, she was sitting at the restaurant	filler	1	1
Filler	The lawyer talked to Peter by the office, the office was quite disorganized.	filler	1	4
Filler	Susan was talking to Mrs. Jones about the patio. Susan was very excited.	filler	1	3
Filler	As Mr. Smith argued with Peter by the box, the box got quite wet.	filler	1	3
Filler	The waiter talked to the lawyer sitting at the patio, it was a very nice patio.	filler	1	2
Filler	The patio was very dirty but Peter and Mr. Smith did not care at all.	filler	1	1
Filler	The plumber was working for Mrs. Jones fixing the fountain, the fountain was broken.	filler	1	4
Filler	Mrs. Jones was talking about work to the lawyer by the hall, the hall was empty.	filler	1	4
Filler	Peter was fighting with Susan about the candy. Peter was angry.	filler	1	3
Filler	While Mrs. Jones was waiting for Peter by the car, the car was running out of gas.	filler	1	2
Filler	The lawyer informed the doctor in the office, the office was a mess.	filler	1	1
Filler	The fountain's water was unsafe, Mrs. Jones and Susan were very unhappy about it.	filler	1	1
Filler	The lawyer was talking to Mrs. Jones by the building, the building was huge.	filler	1	4
Filler	Mr. Smith was happy at the waiter for the nice steak, the nice steak was delicious.	filler	1	3
Filler	Mrs. Jones was paying attention to Peter talk about the garden. Peter was very passionate.	filler	1	2
Filler	Mr. Smith kissed Mrs. Jones in the garden. Immediately afterwards he presented a beautiful ring.	training	1	2
Filler	Peter informed the chef in the kitchen, the kitchen was a mess.	training	1	1
Filler	The lawyer was reimbursing the waiter for the stolen box. The waiter looked weary	training	1	4
SU AM	Peter hugged Mr. Smith by the tree. Immediately afterwards he picked a couple of green leaves.	c	2	3
SU UN	Peter helped Mrs. Jones by the table. Immediately afterwards he ate a couple of nice steaks.	d	2	3
OB AM	Mr. Smith thanked Peter by the car. Immediately afterwards he held a couple of metal nuts.	a	2	2
OB UN	Mrs. Jones brought Peter to the museum. Immediately afterwards he bought a couple of postcards.	b	2	1
SU AM	Peter helped Mr. Smith by the door. Immediately afterwards he bought a couple of oil cans.	c	2	4
SU UN	Peter helped Mrs. Jones by the house. Immediately afterwards he bought a couple of soap bars.	d	2	4
OB AM	Mr. Smith hugged Peter by the stove. Immediately afterwards he ate a couple of sandwiches.	a	2	3
OB UN	Mrs. Jones scolded Peter by the fridge. Immediately afterwards he ate a couple of pancakes.	b	2	2
SU AM	Peter helped Mr. Smith in the office. Immediately afterwards he sent a couple of documents.	c	2	1
SU UN	Peter found Mrs. Jones by the path. Immediately afterwards he drank a couple of glasses of water.	d	2	1
OB AM	Mr. Smith advised Peter by the tent. Immediately afterwards he bought a couple of oranges.	a	2	4
OB UN	Mrs. Jones helped Peter by the shop. Immediately afterwards he chose a couple of rugby balls.	b	2	3
SU AM	Peter scolded Mr. Smith by the sink. Immediately afterwards he cleaned a couple of forks.	c	2	2
SU UN	Peter found Mrs. Jones at the bus stop. Immediately afterwards he bought a couple of tickets.	d	2	2
OB AM	Mr. Smith thanked Peter by the factory. Immediately afterwards he fixed a couple of lightbulbs.	a	2	1
OB UN	Mrs. Jones hugged Peter by the garden. Immediately afterwards he picked a couple of flowers.	b	2	4
Filler	Before Peter talked to Susan by the kitchen, the kitchen was quite dirty.	filler	2	3
Filler	The doctor saw the waiter working in the restaurant, it was a nice place.	filler	2	3
Filler	The restaurant was closing but Susan was happily talking to Mr. Smith.	filler	2	2
Filler	Peter was talking to the doctor by the fountain, the fountain was pretty.	filler	2	1
Filler	The waiter was serving the food to Mrs. Jones by the garden, the garden was nice.	filler	2	4
Filler	Mr. Smith was paying the plumber for fixing the pipes. The plumber was tired.	filler	2	1

Filler	After Susan helped Mr. Smith by the shop, the shop was quite organized.	filler	2	2
Filler	The plumber helped the waiter carry the wheelbarrow, it was quite heavy.	filler	2	3
Filler	The wheelbarrow was very heavy but Mrs. Jones was helping Peter .	filler	2	4
Filler	Susan was asking the waiter for some ice cold water, she was sitting at the restaurant	filler	2	1
Filler	The lawyer talked to Peter by the office, the office was quite disorganized.	filler	2	2
Filler	Susan was talking to Mrs. Jones about the patio. Susan was very excited.	filler	2	3
Filler	As Mr. Smith argued with Peter by the box, the box got quite wet.	filler	2	4
Filler	The waiter talked to the lawyer sitting at the patio, it was a very nice patio.	filler	2	1
Filler	The patio was very dirty but Peter and Mr. Smith did not care at all.	filler	2	2
Filler	The plumber was working for Mrs. Jones fixing the fountain, the fountain was broken.	filler	2	3
Filler	Mrs. Jones was talking about work to the lawyer by the hall, the hall was empty.	filler	2	4
Filler	Peter was fighting with Susan about the candy. Peter was angry.	filler	2	1
Filler	While Mrs. Jones was waiting for Peter by the car, the car was running out of gas.	filler	2	2
Filler	The lawyer informed the doctor in the office, the office was a mess.	filler	2	3
Filler	The fountain's water was unsafe, Mrs. Jones and Susan were very unhappy about it.	filler	2	4
Filler	The lawyer was talking to Mrs. Jones by the building, the building was huge.	filler	2	1
Filler	Mr. Smith was happy at the waiter for the nice steak, the nice steak was delicious.	filler	2	2
Filler	Mrs. Jones was paying attention to Peter talk about the garden. Peter was very passionate.	filler	2	3
Filler	Mr. Smith kissed Mrs. Jones in the garden. Immediately afterwards he presented a beautiful ring.	training	2	4
Filler	Peter informed the chef in the kitchen, the kitchen was a mess.	training	2	1
Filler	The lawyer was reimbursing the waiter for the stolen box. The waiter looked weary	training	2	2
OB UN	Mrs. Jones hugged Peter by the tree. Immediately afterwards he picked a couple of green leaves.	b	3	3
SU AM	Peter helped Mr. Smith by the table. Immediately afterwards he ate a couple of nice steaks.	c	3	4
SU UN	Peter thanked Mrs. Jones by the car. Immediately afterwards he held a couple of metal nuts.	d	3	1
OB AM	Mr. Smith brought Peter to the museum. Immediately afterwards he bought a couple of postcards.	a	3	2
OB UN	Mrs. Jones helped Peter by the door. Immediately afterwards he bought a couple of oil cans.	b	3	3
SU AM	Peter helped Mr. Smith by the house. Immediately afterwards he bought a couple of soap bars.	c	3	4
SU UN	Peter hugged Mrs. Jones by the stove. Immediately afterwards he ate a couple of sandwiches.	d	3	1
OB AM	Mr. Smith scolded Peter by the fridge. Immediately afterwards he ate a couple of pancakes.	a	3	2
OB UN	Mrs. Jones helped Peter in the office. Immediately afterwards he sent a couple of documents.	b	3	3
SU AM	Peter found Mr. Smith by the path. Immediately afterwards he drank a couple of glasses of water.	c	3	4
SU UN	Peter advised Mrs. Jones by the tent. Immediately afterwards he bought a couple of oranges.	d	3	1
OB AM	Mr. Smith helped Peter by the shop. Immediately afterwards he chose a couple of rugby balls.	a	3	2
OB UN	Mrs. Jones scolded Peter by the sink. Immediately afterwards he cleaned a couple of forks.	b	3	3
SU AM	Peter found Mr. Smith at the bus stop. Immediately afterwards he bought a couple of tickets.	c	3	4
SU UN	Peter thanked Mrs. Jones by the factory. Immediately afterwards he fixed a couple of lightbulbs.	d	3	1
OB AM	Mr. Smith hugged Peter by the garden. Immediately afterwards he picked a couple of flowers.	a	3	2
Filler	Before Peter talked to Susan by the kitchen, the kitchen was quite dirty.	filler	3	3
Filler	The doctor saw the waiter working in the restaurant, it was a nice place.	filler	3	4
Filler	The restaurant was closing but Susan was happily talking to Mr. Smith.	filler	3	1
Filler	Peter was talking to the doctor by the fountain, the fountain was pretty.	filler	3	2
Filler	The waiter was serving the food to Mrs. Jones by the garden, the garden was nice.	filler	3	3
Filler	Mr. Smith was paying the plumber for fixing the pipes. The plumber was tired.	filler	3	4
Filler	After Susan helped Mr. Smith by the shop, the shop was quite organized.	filler	3	1
Filler	The plumber helped the waiter carry the wheelbarrow, it was quite heavy.	filler	3	2
Filler	The wheelbarrow was very heavy but Mrs. Jones was helping Peter .	filler	3	3
Filler	Susan was asking the waiter for some ice cold water, she was sitting at the restaurant	filler	3	4
Filler	The lawyer talked to Peter by the office, the office was quite disorganized.	filler	3	1
Filler	Susan was talking to Mrs. Jones about the patio. Susan was very excited.	filler	3	2
Filler	As Mr. Smith argued with Peter by the box, the box got quite wet.	filler	3	3
Filler	The waiter talked to the lawyer sitting at the patio, it was a very nice patio.	filler	3	4
Filler	The patio was very dirty but Peter and Mr. Smith did not care at all.	filler	3	1
Filler	The plumber was working for Mrs. Jones fixing the fountain, the fountain was broken.	filler	3	2
Filler	Mrs. Jones was talking about work to the lawyer by the hall, the hall was empty.	filler	3	3
Filler	Peter was fighting with Susan about the candy. Peter was angry.	filler	3	4
Filler	While Mrs. Jones was waiting for Peter by the car, the car was running out of gas.	filler	3	1
Filler	The lawyer informed the doctor in the office, the office was a mess.	filler	3	2
Filler	The fountain's water was unsafe, Mrs. Jones and Susan were very unhappy about it.	filler	3	3
Filler	The lawyer was talking to Mrs. Jones by the building, the building was huge.	filler	3	4
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The images used for the experiment will be uploaded to this [repository](#).